# Phase-Margin Certification of Multi-Agent LLM Systems: A Behavioral Frequency-Response Approach to Loop-Stability Diagnosis

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April 28, 2026

**Abstract**

Multi-agent large-language-model (LLM) systems have entered production yet fail empirically between 41% and 86.7% of the time on standard benchmarks (Cemri et al., 2025). A substantial fraction of those failures are not reasoning errors but *closed-loop dynamical failures*: agents oscillate, repeat steps, drift into echo-chamber consensus, or run away from themselves until their iteration budget is exhausted. The Multi-Agent System failure Taxonomy (MAST) catalogues fourteen named failure modes across 1,600+ traces with inter-annotator $\kappa = 0.88$, and several explicitly correspond to instabilities of the agent–environment feedback loop. Yet there exists no pre-flight *diagnostic* that would predict, before a costly run, whether a given LLM + prompt + tool + loop will converge, oscillate, or diverge.

We propose such a diagnostic by lifting the recently developed *phase theory of MIMO LTI systems* (Chen et al., 2023b; Wang et al., 2024) to the sampled-time, stochastic, behavioural setting required by black-box LLM agents. We define a *behavioural phase response* of an LLM agent, identifiable from a small number of carefully designed probe perturbations in semantic-embedding space, and prove a *Sampled Behavioural Small-Phase Theorem* that converts the identified phase response into a quantitative phase margin. The margin classifies the agentic loop into one of three regimes that match the empirical taxonomy of Tacheny (2025) and the limit-cycle observations of Wang et al. (2025): *contractive* (will converge), *oscillatory* (will cycle), or *exploratory* (will diverge). The diagnostic returns a single number from $\mathcal{O}(10^2)$ rollouts and addresses approximately six of the fourteen MAST failure categories without access to model weights, attention activations, or training data.

This paper presents the formalism, the theorem, the identification protocol, the mapping to documented failure modes, and an honest catalogue of open problems. We argue this is the first control-theoretic instrument that turns the loop-stability fraction of multi-agent LLM failure from a post-hoc surprise into a measurable margin.

**Keywords.** Multi-agent LLM systems, phase theory, negative imaginary systems, behavioural systems, frequency-response identification, loop stability, agentic AI, robustness certification.

## 1 Introduction

The deployment of multi-agent LLM systems—ReAct-style agents, multi-LLM debate, agentic coding assistants, autonomous research aides—has moved faster than the diagnostics that should accompany any feedback system at scale. This paper is about closing one specific gap in those diagnostics.

### 1.1 The empirical problem

Multi-agent LLM frameworks (AutoGen, MetaGPT, ChatDev, HyperAgent, AppWorld, AG2, Magentic-One) fail empirically between 41% and 86.7% of the time on standard benchmarks, depending on the framework and the task (Cemri et al., 2025). A peer-reviewed study at NeurIPS 2025 catalogues 1,600+ annotated execution traces into 14 named failure modes clustered in 3 categories (*system design*, *inter-agent misalignment*, *task verification*), and validates the taxonomy with inter-annotator agreement $\kappa = 0.88$. Two of the named modes, `step_repetition` and `reasoning_action_mismatch`, correspond directly to closed-loop dynamical failures: oscillation between states and phase decoupling between sub-blocks of the agentic loop. Independent work has reproduced explicit limit cycles in iterated paraphrasing (Wang et al., 2025) and has classified agentic-loop trajectories into contractive, oscillatory, and exploratory dynamical regimes (Tacheny, 2025). In multi-agent debate specifically, homogeneous-agent panels collapse into echo-chamber consensus (Oh et al., 2025), a single adversarial agent can shift system accuracy by 10%–40% (Anonymous Authors, 2026b), and current debate methods fail to consistently improve over single-agent baselines (Anonymous Authors, 2025). Production frameworks have responded by shipping emergency hard-stops such as AutoGen's `max_consecutive_auto_reply`, an implicit acknowledgment that the loop-stability problem is real enough to require an explicit escape hatch.

### 1.2 The diagnostic gap

Despite the documented loss surface, no *pre-flight* diagnostic predicts loop stability for a given LLM, prompt, and tool configuration. Practitioners detect failure post-hoc when the iteration cap is reached, which is the most expensive way to discover divergence. The folklore ("this prompt is unstable", "this loop tends to thrash") is plausible but unmeasured.

In every other engineering discipline that puts feedback loops in front of users—audio, aviation, power grids, automotive—a stability margin is identified once and read out before deployment. The instrument behind that identification is a *frequency-response measurement*: small, structured perturbations applied at the input, the output is measured, gain and phase are extracted, and a margin is computed. The instrument is over a century old in audio (Bode, 1945); it has been generalised to multi-input multi-output systems (Anderson and Vongpanitlerd, 1973; Megretski and Rantzer, 1997; Iwasaki and Hara, 2005), and most recently to phase-bounded systems (Chen et al., 2023b). LLM agentic loops are the first widely-deployed feedback system that has not yet been instrumented with this kind of margin.

### 1.3 Contributions

This paper proposes such an instrument and develops the supporting theory. Specifically:

1. We model the LLM agentic loop as a sampled, nonlinear, stochastic feedback system on a finite-dimensional semantic-embedding manifold (Section 4).

2. We introduce the *behavioural phase response* of an LLM agent: an empirical phase identifiable from a small number of probe perturbations of the agent's nominal trajectory in semantic-embedding space, without access to weights or attention activations (Section 5).

3. We define a corresponding *locally sectorial* class of LLM agents—the behavioural analogue of the (semi-)sectorial systems of Chen et al. (2023b)—and prove a *Sampled Behavioural Small-Phase Theorem* that converts the identified phase response into a quantitative loop-stability margin $\Phi_{\text{margin}}$ (Section 6).

4. We give a black-box probe protocol that produces $\Phi_{\text{margin}}$ from $\mathcal{O}(10^2)$ rollouts (Section 7), and predict the three regimes (contractive / oscillatory / exploratory) reported in the empirical literature (Section 6.1).

5. We map the certificate explicitly onto roughly six of the fourteen MAST failure modes, and we are explicit about which modes the certificate does *not* address (Section 8).

6. We propose an experimental validation plan against the published benchmarks and the documented limit-cycle datasets (Section 9), and we list four open theoretical problems that the framework forces (Section 10).

The novelty claim is precise. The phase theory of LTI systems is in print (Chen et al., 2023b); its nonlinear extension is in print (Chen et al., 2021); the empirical taxonomy of multi-agent LLM failure is in print (Cemri et al., 2025); the geometric description of agentic loop dynamics is in print (Tacheny, 2025). What is *not* in print, to the best of our literature audit, is a *sampled-time, stochastic, behavioural* phase certificate identified from black-box LLM probes that produces a quantitative margin and matches it to the documented failure modes. That is what we contribute.

# 2 The Documented Problem

We begin by establishing, with citations, that the loop-stability fraction of multi-agent LLM failures is real, named, measured, and currently without a pre-flight diagnostic.

## 2.1 The MAST taxonomy and quantitative failure rates

Cemri et al. (2025) construct the first multi-agent system failure taxonomy, MAST, by annotating 1,600+ execution traces collected from seven popular multi-agent LLM frameworks across coding, math, and general agent tasks. The headline rate is the empirical task-failure rate per framework, which ranges from 41% to 86.7% on standard benchmarks. The fourteen failure modes are organised as follows.

- *System design issues:* disobeying task specification, disobeying role specification, step repetition, loss of conversational history, premature termination.
- *Inter-agent misalignment:* reasoning-action mismatch, information withholding, ignored other agent's input, action–reasoning disagreement, derailing, withholding critical information, incorrect verification.
- *Task verification:* insufficient verification, incorrect verification of action outputs.

The two italicised modes `step_repetition` and `reasoning_action_mismatch` correspond directly to feedback-loop failures. Together with the qualitative descriptions in the paper of "conversations cycling without progress", "perpetual cycle of unproductive exchanges", and "memory limits causing agents to forget previous discussions" (op. cit.), they constitute the loop-stability fraction of the failure surface.

## 2.2 Independent reproduction of the dynamical regimes

Two recent dynamical-systems analyses corroborate the picture. Tacheny (2025) formalise the LLM agentic loop as a discrete-time dynamical system in semantic-embedding space and exhibit *three classifiable regimes*: contractive trajectories converging to stable attractors, oscillatory trajectories cycling among attractors, and exploratory trajectories diverging unboundedly. Wang et al. (2025) demonstrate an explicit *2-period limit cycle* in iterated paraphrasing across multiple

frontier LLMs—a textbook oscillation that would, in any other engineering discipline, be diagnosed by a phase margin.

### 2.3 Multi-agent debate failure

The picture sharpens for multi-agent debate, the prototypical multi-agent LLM use case. Oh et al. (2025) document that homogeneous-agent debate panels frequently converge into echo-chamber consensus that amplifies pre-existing biases instead of correcting them, with the effect intensifying as panel size grows. In sequential strategic environments, a single early suboptimal decision triggers a cascade that the consensus process cannot self-correct. Anonymous Authors (2026b) quantify adversarial influence: a single strategically designed adversarial agent reduces system accuracy by 10%–40% and increases consensus on incorrect answers by more than 30% on factual benchmarks. Wynn et al. (2025) catalogue further debate failure modes including stalemate, premature agreement, and degenerate reasoning. The synthesis of Anonymous Authors (2025) concludes that current multi-agent debate methods do not reliably outperform single-agent baselines, even with substantially increased compute—and that simple majority voting captures most of whatever benefit accrues.

### 2.4 Production-side acknowledgement

The folklore from production deployments matches the academic findings. The popular AutoGen framework added an explicit `max_consecutive_auto_reply` parameter as an emergency hard-stop; several agent-deployment vendors have published essays titled "the agent loop problem" or "stop the loop". The major LLM providers all impose default iteration caps on their agent products—an implicit confession that the underlying systems do not converge unaided.

### 2.5 The diagnostic gap

In every cited reference above the failure modes are detected *after* the loop has run. No published method takes an LLM, a prompt, a tool stack, and a loop topology as input and returns a quantitative pre-flight stability margin as output. The *description* of the regimes is in print; the *prediction instrument* is not. Filling this gap is the goal of the present paper.

## 3 Control-Theoretic Background

We summarise, at the level of working definitions, the four control-theoretic ingredients we draw on: positive realness and passivity, negative imaginary systems, the phase theory of Chen et al. (2023b), and the behavioural systems framework of Willems (1986).

### 3.1 Positive realness and passivity

A real-rational proper transfer matrix $G : \mathbb{C} \to \mathbb{C}^{m\times m}$ is *positive real* (PR) if it is analytic in $\Re s > 0$ and $G(s) + G(s)^* \succeq 0$ for $\Re s > 0$. By the Kalman–Yakubovich–Popov (KYP) lemma (Anderson and Vongpanitlerd, 1973) this is equivalent to the existence of $P = P^\top \succ 0$ certifying a state-space dissipation inequality with supply rate $s(u, y) = u^\top y$. Two PR systems in negative feedback are stable under mild non-degeneracy, the original "passivity theorem".

### 3.2 Negative imaginary systems

A real-rational proper $G$ is *negative imaginary* (NI) (Lanzon and Petersen, 2008; Mabrok et al., 2014) if it is analytic in $\Re s > 0$ and $j(G(j\omega) - G(j\omega)^*) \succeq 0$ for all $\omega > 0$ where $j\omega$ is not a pole, equivalently $\Im G(j\omega) \leq 0$. NI captures lightly damped flexible-structure plants (collocated

force–position) for which PR fails. Two NI systems in positive feedback are stable under a DC-gain condition $\lambda_{\max}(M(0)N(0)) < 1$.

### 3.3 Phase theory of MIMO LTI systems

Chen et al. (2023b) unify PR and NI into a single phase-bounded class. Briefly, a square real-rational $G$ is *(semi-)sectorial* if $G(j\omega)$ is sectorial as a complex matrix at every frequency, and its *phase response* $\phi(G(j\omega)) \subset (-\pi, \pi]$ is well-defined and lies in a frequency-dependent phase sector. PR corresponds to phase contained in $[-\pi/2, \pi/2]$; NI to phase in $[-\pi, 0]$. The *Small Phase Theorem* of Chen et al. (2023b) states that $[G_1, G_2]_-$ is internally stable if $\sup_\omega(\overline{\phi}(G_1(j\omega)) + \overline{\phi}(G_2(j\omega))) < \pi$ and a non-encirclement condition holds, where $\overline{\phi}$ is the upper phase. An accompanying *Sectored Real Lemma* provides an LMI characterisation, and a nonlinear extension exists for input–output sectorial systems (Chen et al., 2021). Mixed gain–phase robustness is treated in (Zhao et al., 2022).

These results are the foundation we will lift.

### 3.4 Behavioural systems theory

Willems (1986, 2007) pioneered the view that a dynamical system is defined by the set of admissible input–output trajectories ("the behaviour"), independent of any particular state-space realisation. This is the right standpoint for LLM agents, whose internal mechanism (weights, attention activations) is typically inaccessible and irrelevant for closed-loop stability: what matters is the input–output map at the embedding level. The data-driven realisation of behavioural ideas in DeePC (Coulson et al., 2019) and the fundamental lemma of Willems et al. (2005) demonstrate that meaningful control-theoretic objects can be extracted from data alone.

## 4 The LLM Agentic Loop as a Sampled Feedback System

We now formalise the object the certificate will measure.

### 4.1 State-space in semantic-embedding coordinates

Fix a frozen embedding map $\Pi : \Sigma^* \to \mathcal{Z} \subset \mathbb{R}^d$ from token sequences to a finite-dimensional semantic vector space (e.g. sentence-T5, the LLM's own pre-LM-head residual stream, or any fixed-vocabulary representation). Let $z_k \in \mathcal{Z}$ denote the embedded running context at iteration $k$ of an agentic loop, and $y_k \in \mathbb{R}^p$ a measurable observable extracted from $z_k$ via a linear or smooth nonlinear projection $\mathcal{H} : \mathcal{Z} \to \mathbb{R}^p$ (e.g. a goal-direction projection, a tool-call type indicator, or a probe class label).

**Assumption 4.1** (Embedding regularity)**.** The embedding map $\Pi$ and the observation map $\mathcal{H}$ are $L$-Lipschitz on a neighbourhood of the nominal trajectory $\{z_k^\star\}_k$ used to compute probes.

### 4.2 Closed-loop dynamics

The agentic loop is the sampled-time discrete dynamical system

$$z_{k+1} = \mathcal{L}_\theta(z_k,\, u_k,\, \xi_k), \qquad y_k = \mathcal{H}(z_k), \tag{1}$$

$$u_{k+1} = \mathcal{E}(y_k,\, w_k), \tag{2}$$

where $\mathcal{L}_\theta : \mathcal{Z} \times \mathbb{R}^m \times \mathbb{R}^q \to \mathcal{Z}$ is the LLM embedded in the loop topology (decoder + tool router + memory), $\mathcal{E} : \mathbb{R}^p \times \mathbb{R}^r \to \mathbb{R}^m$ is the environment / tool block (possibly itself another LLM), $\xi_k$ collects the LLM's stochastic decoding sources, and $w_k \sim P_w$ is exogenous environmental noise.

The closed-loop map is

$$z_{k+1} = \Phi(z_k, z_{k-1}, \xi_k, w_k) := \mathcal{L}_\theta\Big(z_k,\, \mathcal{E}\big(\mathcal{H}(z_{k-1}), w_k\big),\, \xi_k\Big). \tag{3}$$

Equation (3) is a discrete-time, nonlinear, stochastic dynamical system on $\mathcal{Z} \subset \mathbb{R}^d$.

### 4.3 Why existing phase theory does not directly apply

The phase theory of Chen et al. (2023b) is LTI and continuous-time; its nonlinear extension (Chen et al., 2021) is deterministic. The map $\mathcal{L}_\theta$ in (1) is

(i) **Discrete-time and sampled.** The LLM is queried at integer iterations; the natural frequency variable is $\Omega \in (0, \pi]$ in samples-per-iteration units.

(ii) **Nonlinear and not analytically known.** $\mathcal{L}_\theta$ is not expressible in closed form; only black-box rollouts are available.

(iii) **Stochastic.** Decoding produces output variance even with identical input; $\xi_k$ contributes a non-trivial $\sigma_w^2$.

(iv) **Time-varying along the trajectory.** $\mathcal{L}_\theta$'s effective linearisation depends on the running context $z_k^\star$; the behaviour is at best *locally* Linear Parameter-Varying.

(v) **Defined on a high-dimensional embedding manifold.** Phase must be defined directionally with respect to a probing basis in $\mathcal{Z}$.

These five features force a behavioural, sampled, stochastic generalisation of the small-phase theorem. Sections 5 and 6 construct that generalisation.

## 5 Behavioural Phase Response of an LLM Agent

We define the phase response of $\mathcal{L}_\theta$ not by differentiation of an analytic model but by observation of input–output behaviour around a nominal trajectory.

### 5.1 Probe protocol input

Fix a nominal trajectory $\{(z_k^\star, u_k^\star)\}_{k=0}^{N-1}$ obtained by running the loop on a representative task. Pick a unit perturbation direction $v \in \mathcal{Z}$, $\|v\| = 1$, a probe amplitude $\epsilon > 0$, and a probe discrete frequency $\Omega \in (0, \pi]$. Inject the perturbation at the agent's input,

$$u_k(\Omega, v) = u_k^\star + \epsilon\, v \cos(\Omega k), \qquad k = 0, \ldots, N-1, \tag{4}$$

and run $M$ independent stochastic rollouts of the open-loop agent $\mathcal{L}_\theta$ (i.e. with $\mathcal{E}$ replaced by the deterministic playback of $u_k(\Omega, v)$) of length $N$ each. Let $\bar{y}_k(\Omega, v) := \frac{1}{M}\sum_{m=1}^{M} \mathcal{H}(z_k^{(m)})$ be the seed-averaged observable.

### 5.2 Empirical phase identification

Define the residual against the nominal, $\Delta y_k(\Omega, v) := \bar{y}_k(\Omega, v) - \mathcal{H}(z_k^\star)$, and fit a sinusoidal model with phase $\theta$, amplitude $A$:

$$\big(\widehat{\theta}_v(\Omega),\, \widehat{A}_v(\Omega)\big) := \underset{\theta \in (-\pi, \pi],\, A \geq 0}{\arg\min} \sum_{k=0}^{N-1} \Big\| \Delta y_k(\Omega, v) - \epsilon\, A \cos(\Omega k + \theta)\, b(v) \Big\|_2^2, \tag{5}$$

where $b(v) := \partial\mathcal{H}/\partial z\big|_{z_k^\star} v \in \mathbb{R}^p$ is the probe response direction in observable space (precomputed once via finite differences if $\mathcal{H}$ is not given). Equation (5) is a complex linear regression, solvable in closed form.

**Definition 5.1** (Behavioural phase of an LLM agent)**.** The *behavioural phase response* of $\mathcal{L}_\theta$ around $\{z_k^\star\}$ in direction $v$ at frequency $\Omega$ is $\widehat{\theta}_v(\Omega) \in (-\pi, \pi]$ obtained from (5). We similarly define $\widehat{\phi}_v(\Omega)$ for the environment $\mathcal{E}$ by perturbing $y_{k-1}$ and observing the induced perturbation in $u_k$.

### 5.3 Locally sectorial agents

**Definition 5.2** (Locally sectorial agent)**.** The agent $\mathcal{L}_\theta$ is *locally sectorial of class* $\mathcal{S}(\alpha, \beta)$ around $\{z_k^\star\}$ in direction $v$ at fit residual level $\eta > 0$ if for every $\Omega \in (0, \pi]$,

$$\widehat{\theta}_v(\Omega) \in [\alpha, \beta], \tag{6}$$

and the relative residual $\rho_v(\Omega) := \frac{\sum_k \|\Delta y_k(\Omega,v) - \epsilon \widehat{A}_v(\Omega)\cos(\Omega k + \widehat{\theta}_v(\Omega))b(v)\|^2}{\sum_k \|\Delta y_k(\Omega,v)\|^2} \leq \eta$.

*Remark* 5.3. If $\mathcal{L}_\theta$ happens to be LTI with transfer matrix $G$ on the linearisation around $\{z_k^\star\}$ in direction $v$, then for $\epsilon \to 0$, $M \to \infty$, $\sigma_w \to 0$ we recover $\widehat{\theta}_v(\Omega) \to \arg(b(v)^\top G(e^{j\Omega})v)$, the standard phase response. Definition 5.1 is therefore a behavioural generalisation that reduces correctly.

*Remark* 5.4. The phase response in Definition 5.1 is direction-dependent because $\mathcal{L}_\theta$ is a high-dimensional MIMO operator. In practice we identify it on a probing basis $\{v_1, \ldots, v_p\}$ chosen via the top-$p$ principal directions of the gradient of $\mathcal{H}$ along the nominal trajectory, or via task-relevant semantic axes (e.g. answer-stance, plan specificity, tool-call category). See Section 7.2.

### 5.4 Sample-complexity of the identification

**Lemma 5.5** (Phase identification rate)**.** *Under Assumption 4.1 and bounded decoding noise* $\mathbb{E}[\|\xi_k\|^2] \leq \sigma_w^2$*, suppose the local linearisation of* $\mathcal{L}_\theta$ *in direction* $v$ *at frequency* $\Omega$ *has gain* $A_v^\star(\Omega) \geq A_{\min} > 0$ *and phase* $\theta_v^\star(\Omega)$*. Then the empirical phase* $\widehat{\theta}_v(\Omega)$ *from* (5) *satisfies*

$$\mathbb{E}\big|\widehat{\theta}_v(\Omega) - \theta_v^\star(\Omega)\big|^2 \;\leq\; \frac{c\,\sigma_w^2}{\epsilon^2\, A_{\min}^2\, M\, N} + \mathcal{O}(\eta),$$

*for an absolute constant* $c$*, with the second term controlling the local-linearity bias.*

The lemma is a standard sinusoid-in-Gaussian-noise least-squares calculation; the proof is sketched in Appendix A. The headline takeaway is that doubling the probe amplitude or the sample budget halves the phase identification variance, and the local-linearity bias is $\mathcal{O}(\eta)$ rather than $\mathcal{O}(\eta^2)$, so the choice of probe amplitude is a bias-variance trade.

## 6 The Sampled Behavioural Small-Phase Theorem

We now state the main theorem. Let $W_1$ denote the Wasserstein-1 distance between probability measures on $\mathcal{Z}$.

**Assumption 6.1** (Local linearity envelope)**.** The agent and environment maps $\mathcal{L}_\theta$ and $\mathcal{E}$ admit local linearisations along the nominal trajectory whose Jacobian operator norms in the probing basis are bounded above by $L_{\mathcal{L}_\theta}$ and $L_{\mathcal{E}}$, with residuals bounded by $\eta_{\mathcal{L}_\theta}$ and $\eta_{\mathcal{E}}$ in the sense of Definition 5.2.

**Assumption 6.2** (Stochastic non-degeneracy)**.** The decoding noise $\xi_k$ has a density bounded above and below on its support, and the map $\mathcal{L}_\theta$ is irreducible in the sense of having no absorbing region of zero measure.

**Theorem 6.3** (Sampled Behavioural Small-Phase)**.** *Suppose, around the nominal trajectory and in every direction $v$ of a probing basis $\{v_1, \dots, v_p\}$, the agent and environment satisfy*

$$\mathcal{L}_\theta \in \mathcal{S}(\alpha_v, \beta_v), \qquad \mathcal{E} \in \mathcal{S}(\gamma_v, \delta_v), \qquad v \in \{v_1, \dots, v_p\}. \tag{7}$$

*Define the directional* phase margin

$$\Phi_{\text{margin}}(v) := \min\big(\, \pi - (\beta_v + \delta_v),\ (\alpha_v + \gamma_v) + \pi \,\big), \tag{8}$$

*and the global margin $\Phi_{\text{margin}} := \min_{v \in \{v_1, \dots, v_p\}} \Phi_{\text{margin}}(v)$. Under Assumptions 4.1–6.2, if*

$$\Phi_{\text{margin}} \ > \ C(\eta_{\mathcal{L}_\theta} + \eta_{\mathcal{E}}) \ + \ \frac{C' \sigma_w}{\epsilon\, A_{\min}} \frac{1}{\sqrt{MN}}, \tag{9}$$

*for explicit constants $C, C' > 0$ depending only on $L_{\mathcal{L}_\theta}, L_{\mathcal{E}}, L, d, p$, then the closed-loop process $\{z_k\}$ defined by (3) is*

*(1) geometrically ergodic with a unique invariant measure $\pi^\infty$,*

*(2) Wasserstein-1 contractive at rate $\rho \le 1 - \Phi_{\text{margin}}/\pi + \mathcal{O}(\eta_{\mathcal{L}_\theta} + \eta_{\mathcal{E}} + (MN)^{-1/2})$, in the sense that $W_1(\mathcal{P}_k \mu, \pi^\infty) \le \rho^k W_1(\mu, \pi^\infty)$ for any initial distribution $\mu$,*

*(3) the empirical observable $y_k$ converges in distribution to a unique stationary law and the running mean settles within $\varepsilon$ of its asymptotic value in $\mathcal{O}(\frac{1}{\Phi_{\text{margin}}/\pi} \log \frac{1}{\varepsilon})$ iterations.*

A proof sketch of Theorem 6.3 is given in Appendix A. The strategy combines three ingredients: the deterministic small-phase argument of (Chen et al., 2023b, 2021) bounding the deterministic part of the closed-loop iteration, a local-linearity perturbation bound paying $\mathcal{O}(\eta_{\mathcal{L}_\theta} + \eta_{\mathcal{E}})$, and a discrete-time stochastic Lyapunov / drift argument paying $\mathcal{O}((MN)^{-1/2})$ for the empirical-margin estimation error.

### 6.1 Three regimes from one number

The contrapositive of Theorem 6.3 produces a regime classification that matches the empirical literature (Tacheny, 2025; Wang et al., 2025; Cemri et al., 2025).

**Corollary 6.4** (Regime classification)**.** *With $\Phi_{\text{margin}}$ defined as in (8), the agentic loop exhibits*

- contractive *dynamics (predicted convergence to a unique attractor) when $\Phi_{\text{margin}} > 0$ with sufficient confidence;*
- oscillatory *dynamics (predicted limit cycle, the Wang et al. (2025) regime) when $\Phi_{\text{margin}} \approx 0$ in expectation but its directional minimum is negative on a positive-measure set of probe directions;*
- exploratory / divergent *dynamics (predicted unbounded drift) when $\alpha_v + \gamma_v < -\pi$ or $\beta_v + \delta_v > \pi$ on a positive-measure direction set.*

The three regimes are exactly the three observed in Tacheny (2025), but here they are predicted from a quantitative margin computed before the loop is run.

### 6.2 Specialisation to known cases

- *Deterministic LTI agent.* Setting $\sigma_w = 0$, $\eta_{\mathcal{L}_\theta} = \eta_{\mathcal{E}} = 0$, $M = N = \infty$, the residual term in (9) vanishes and Theorem 6.3 reduces to the small-phase theorem of Chen et al. (2023b).
- *Deterministic nonlinear agent.* Setting $\sigma_w = 0$ but keeping $\eta > 0$, we recover a discrete-time analogue of the nonlinear small-phase theorem of Chen et al. (2021).
- *Memoryless map* ($\mathcal{L}_\theta$ and $\mathcal{E}$ both phase-zero around their fixed point). $\Phi_{\text{margin}}$ collapses to a small-gain condition; the iteration is contractive iff $L_{\mathcal{L}_\theta} L_{\mathcal{E}} < 1$, which is the Banach (1922) fixed-point principle.

**Algorithm 1: Black-box phase-margin certification of an LLM agentic loop.**

**Input:** LLM API $\mathcal{L}_\theta$, environment / tool block $\mathcal{E}$, observable $\mathcal{H}$, nominal task prompt $p$, probing basis $V = \{v_1, \ldots, v_p\} \subset \mathcal{Z}$, frequency grid $\mathcal{G} \subset (0, \pi]$, probe amplitude $\epsilon$, sample budget $M$, horizon $N$.

**1.** Run $M_0$ nominal closed-loop rollouts of length $N$ and compute $\{z_k^\star\}$ as the running mean trajectory.

**2.** **For** each $v \in V$ and each $\Omega \in \mathcal{G}$ **do**:

- **a.** Inject $u_k(\Omega, v) = u_k^\star + \epsilon v \cos(\Omega k)$ in the open-loop agent for $M$ stochastic rollouts of horizon $N$.
- **b.** Compute residuals $\Delta y_k(\Omega, v)$ and fit $(\widehat{\theta}_v(\Omega), \widehat{A}_v(\Omega))$ via (5).
- **c.** Compute fit quality $\rho_v(\Omega)$. If $\rho_v(\Omega) > \eta$ for many $\Omega$, expand the probing basis or reduce $\epsilon$.

**3.** Repeat steps 2–3 for $\mathcal{E}$ to obtain $\widehat{\phi}_v(\Omega)$.

**4.** Compute $\alpha_v, \beta_v, \gamma_v, \delta_v$ as min / max over $\Omega \in \mathcal{G}$ with statistical buffers from Lemma 5.5.

**5.** Compute $\Phi_{\text{margin}}(v)$ via (8) and $\Phi_{\text{margin}} = \min_v \Phi_{\text{margin}}(v)$.

**Output:** $\Phi_{\text{margin}}$ and the predicted regime (contractive / oscillatory / exploratory) from Corollary 6.4.

# 7 The Black-Box Probe Protocol

The theorem becomes useful when its hypothesis is checkable from black-box LLM rollouts. Algorithm 1 below gives the probe protocol.

## 7.1 Algorithm

## 7.2 Probing basis selection

The choice of $V = \{v_1, \ldots, v_p\}$ trades off coverage against probe budget. Three natural defaults:

1. *Task-direction basis.* Use a small set of semantic axes relevant to the task: answer stance ($v_{\text{stance}}$), plan specificity ($v_{\text{plan}}$), tool-call category ($v_{\text{tool}}$). Suitable when domain knowledge of the task is available.

2. *Empirical PCA basis.* Run nominal rollouts, compute the PCA of the observed $\{\Delta\mathcal{H}(z_k^\star)\}_k$ residuals against a slightly perturbed prompt, take the top $p$ components. Domain-free default.

3. *Random Hutchinson basis.* Sample $v \sim \text{Unif}(\mathbb{S}^{d-1})$ and use Hutchinson-style estimators to upper-bound $\Phi_{\text{margin}}$ across all directions in expectation. Useful for high-dimensional safety-critical applications.

## 7.3 Computational cost

The probe protocol requires $|V| \cdot |\mathcal{G}| \cdot M \cdot N$ open-loop agent invocations for $\mathcal{L}_\theta$, plus the same for $\mathcal{E}$ if it is not deterministic. With $|V| = 4$, $|\mathcal{G}| = 8$, $M = 8$, $N = 16$, this is $\approx$ 4,000 LLM calls—roughly the cost of two failed agent runs in production—to predict the success of all subsequent runs on the same loop topology.

# 8 Mapping to Documented MAST Failure Modes

We are explicit about which of the fourteen MAST failure modes the phase-margin certificate addresses.

The certificate cleanly addresses approximately *half* of the fourteen MAST modes. This is the loop-stability fraction of the failure surface, which is precisely the fraction with no current pre-flight diagnostic.

| MAST mode | Predicted by $\Phi_{\text{margin}}$? |
|---|---|
| `step_repetition` | **Yes.** Negative directional margin on a stance / plan direction predicts a 2-cycle limit cycle. |
| `reasoning_action_mismatch` | **Yes.** Phase mismatch between sub-block agents shows up as $\beta_v + \delta_v \to \pi$ on the relevant direction. |
| `loss_of_conversation_history` | **Partially.** Memory truncation manifests as a discontinuity in the fit $\widehat{\theta}_v(\Omega)$; flagged but not fully diagnosed. |
| `premature_termination` | **Partially.** Over-contractive margin ($\Phi_{\text{margin}} \to \pi$) flags collapse to a too-quick fixed point. |
| `disobeying_task_specification` | **No.** Specification errors are upstream of the loop dynamics. |
| `disobeying_role_specification` | **No.** Same. |
| `information_withholding` | **No.** Inter-agent communication content; not a loop-stability property. |
| `ignored_other_agent's_input` | **Partially.** Shows up as low directional gain $\widehat{A}_v(\Omega)$, indicating a near-zero loop element in that direction. |
| `action_reasoning_disagreement` | **Yes.** Internal phase mismatch within $\mathcal{L}_\theta$ between reasoning and acting sub-modules. |
| `derailing` | **Partially.** Exploratory regime is predicted by $\alpha_v + \gamma_v < -\pi$. |
| `withholding_critical_information` | **No.** |
| `insufficient_verification` | **No.** Verification quality is a content property. |
| `incorrect_verification` | **No.** |
| `incorrect_verification_of_action_outputs` | **No.** |

Table 1: Mapping the phase-margin certificate to the MAST taxonomy of Cemri et al. (2025). Of the fourteen named modes, the certificate fully addresses three, partially addresses four, and is silent on seven. The seven on which it is silent are content / specification issues, not loop-dynamics issues.

# 9 Validation Plan

The certificate is falsifiable. We propose three validation experiments against published datasets and benchmarks.

## 9.1 Experiment 1: Three-regime classification

Tacheny (2025) provide a benchmark of agentic-loop trajectories labelled into the three dynamical regimes (contractive, oscillatory, exploratory). We propose:

1. Run the probe protocol (Algorithm 1) on each benchmark configuration to obtain $\Phi_{\text{margin}}$.
2. Apply the regime predictor of Corollary 6.4.
3. Evaluate the confusion matrix against the ground-truth regimes.

*Headline metric:* predicted-vs-actual regime accuracy, with the null baseline being the prior class frequencies. Expected lift if the theory is correct: substantial (the regimes are deterministic in $\Phi_{\text{margin}}$ at low noise).

## 9.2 Experiment 2: 2-cycle prediction in successive paraphrasing

Wang et al. (2025) demonstrate explicit 2-period limit cycles in iterated paraphrasing across multiple frontier LLMs. We propose:

1. Run the probe protocol with the paraphrasing prompt held fixed and probe directions chosen along stance / lexical-density axes.

2. Predict 2-cycle vs. fixed-point convergence from the directional sign of $\Phi_{\text{margin}}(v)$.

*Headline metric:* per-prompt cycle-vs-no-cycle prediction accuracy. The published behaviour is well-controlled (specific prompts exhibit cycles, others do not), so this is a clean falsification target.

### 9.3 Experiment 3: MAST step-repetition trace flagging

The MAST dataset (Cemri et al., 2025) releases 1,600+ annotated traces. We propose:

1. For each MAS framework, run the probe protocol on the framework's default loop topology with a held-out task.

2. Predict per-task whether `step_repetition` or `reasoning_action_mismatch` will trigger.

3. Compare against the MAST annotations on the held-out task.

*Headline metric:* F1 score for predicting these two MAST modes, with the baseline being the per-framework base rate.

### 9.4 Cost and feasibility

Each probe run costs $\approx$ 4,000 LLM calls per loop topology, well within an academic budget for the three frontier APIs. All three benchmarks are publicly released with reproducible evaluation scripts.

## 10 Limitations and Open Problems

We surface the genuine difficulties. These are not minor.

**O1: Embedding-space dependence.** $\Phi_{\text{margin}}$ depends on $\Pi$ and on the probing basis $V$. An invariant formulation—e.g. in terms of the information-geometric Fisher metric of the agent's output distribution—would give a coordinate-free margin. This is open.

**O2: Local linearity assumption.** Definition 5.2 requires the residual $\rho_v(\Omega)$ to be small. When the agent's behaviour is sharply nonlinear (e.g. crossing a tool-call decision boundary), the fit residual exceeds $\eta$ and the certificate degrades. A higher-order behavioural phase, e.g. via Volterra-series or kernel-Hilbert-space identification, would extend the framework but is open.

**O3: Variance reduction.** At realistic decoding temperatures $T = 0.7$, $\sigma_w^2$ may eat the margin. Variance-reduction designs (antithetic seeds, common random numbers across frequencies, or deterministic-decode probes followed by stochastic-decode replays) are needed to make the protocol tractable at high temperature.

**O4: Adversarial robustness.** The certificate is established around a nominal trajectory. An adversarial input distribution voids the certificate. The robust-IQC version of phase theory (Megretski and Rantzer, 1997; Zhao et al., 2022) should extend to give a robust phase margin, but the lift to behavioural identification is open.

**O5: Multi-agent extensions.** We have stated the theorem for a two-block loop $[\mathcal{L}_\theta, \mathcal{E}]_-$. Multi-agent debate involves $k > 2$ agents in arbitrary topologies. The cyclic small-phase theorem (Wang et al., 2023a) treats the LTI case; its behavioural-stochastic generalisation is open.

## 11 Related Work

**Phase theory of MIMO LTI systems.** The substrate of (Chen et al., 2023b; Wang et al., 2024; Chen et al., 2021; Zhao et al., 2022; Wang et al., 2022, 2023b,a) is what we lift. Their framework is LTI (or deterministic nonlinear in the input-output sectorial sense); our contribution is the sampled, stochastic, behavioural extension specific to LLM agents.

**Negative imaginary systems.** PR/NI/passivity foundations: (Anderson and Vongpanitlerd, 1973; Lanzon and Petersen, 2008; Mabrok et al., 2014; Patra and Lanzon, 2011; Griggs et al., 2007). The recent neural-ODE application (Mabrok et al., 2025) controls mechanical plants with NN controllers carrying the NI property; our target is loop dynamics of LLM agents, a different object.

**Behavioural and data-driven systems.** Willems (1986, 2007); Willems et al. (2005); Coulson et al. (2019). These motivate the black-box-probe formulation of phase identification.

**Lipschitz / SDP certification of neural networks.** (Fazlyab et al., 2019; Authors of LipShiFT, 2024; Authors of LipNeXt, 2025). These methods constrain magnitude (gain), not phase, and treat the network statically rather than as a closed-loop system.

**Dynamical-systems analyses of LLM loops.** (Tacheny, 2025; Wang et al., 2025). These describe the regimes; we predict them.

**Multi-agent LLM failure analyses.** (Cemri et al., 2025; Oh et al., 2025; Anonymous Authors, 2026b; Wynn et al., 2025; Du et al., 2024; Anonymous Authors, 2025). These document and explain the failures; we add a pre-flight predictor.

**Stable architectures for deep learning.** (Haber and Ruthotto, 2017) construct continuous-time stable ResNets; orthogonal to the loop-stability question we address.

**Score-based generative model stability.** (Chen et al., 2023a; Anonymous Authors, 2026a). Dissipativity for diffusion sampling; orthogonal to LLM agent loops.

## 12 Conclusion

Multi-agent LLM systems have a documented loop-stability problem. Multi-agent system failure rates of 41%–86.7%, a peer-reviewed 14-mode taxonomy, independent reproductions of explicit limit cycles, adversarial-influence accuracy losses of 10–40%, and emergency hard-stops shipped in production frameworks all point at the same conclusion: the closed-loop dynamics of these systems are unstable in real ways and there is no pre-flight diagnostic.

This paper proposes such a diagnostic by lifting the recently developed phase theory of MIMO LTI systems to the sampled-time, stochastic, behavioural setting required by black-box LLM agents. The Sampled Behavioural Small-Phase Theorem converts a small number of probe rollouts into a single number $\Phi_{\text{margin}}$ that predicts which of three regimes the loop will exhibit, and the regime classification matches the empirical taxonomy in the literature.

The certificate addresses approximately half of the fourteen documented MAST failure modes—specifically the loop-stability half, which has no current pre-flight diagnostic. The other half (specification, verification, content) belongs to other categories of analysis and remains out of scope. Five open problems—embedding invariance, local linearity, variance reduction, adversarial robustness, and the multi-agent extension—are honestly catalogued.

The bigger picture: every other engineering discipline that puts feedback systems in front of users has long since had a way to measure stability margins before the user is exposed. Bringing

that instrument over to the LLM agent's bench is the small, specific, useful, and—to the best of our literature audit—genuinely unaddressed contribution.

# A Proof sketches

### Lemma 5.5

The least-squares fit (5) is equivalent to a complex linear regression on the discrete Fourier basis at frequency $\Omega$. Stack the residuals $\Delta y_k$ along $b(v)$ into a complex sufficient statistic $S_N(\Omega) := \sum_{k=0}^{N-1} \Delta y_k^\top b(v)\, e^{-j\Omega k}$. Under decoding noise of variance $\sigma_w^2$ on $\bar{y}_k$ (after seed-averaging by $M$), the noise in $S_N$ is complex Gaussian with variance $N\sigma_w^2 \|b(v)\|^2/M$. The deterministic part is $\frac{\epsilon N}{2} A_v^\star(\Omega) e^{j\theta_v^\star(\Omega)} \|b(v)\|^2 + \mathcal{O}(\eta\epsilon N)$. Phase identification reduces to $\arg(S_N)$, whose variance is controlled by the noise-to-signal ratio in the standard $|\Delta \arg|^2 \le |\text{noise}|^2/|\text{signal}|^2$ inequality (Rife and Boorstyn, 1974), giving the stated rate. The $\mathcal{O}(\eta)$ bias term is the local-linearity residual mass leaking into the wrong frequency bin. □

### Theorem 6.3

The strategy mirrors the proof of the Small Phase Theorem in (Chen et al., 2023b) but lifts each of three pieces.

**(i) Deterministic sectorial composition.** Around the nominal trajectory in any direction $v$, the linearised iteration on the residual is $\widetilde{z}_{k+1} = J_v \widetilde{z}_k$, where $J_v$ is the composition of two locally-linear maps with phase responses in $[\alpha_v, \beta_v]$ and $[\gamma_v, \delta_v]$. The composed phase is in $[\alpha_v+\gamma_v, \beta_v+\delta_v]$. By the deterministic small-phase theorem, the spectral radius $\rho(J_v) \le \cos(\Phi_{\text{margin}}(v)/2) < 1$ if $\Phi_{\text{margin}}(v) > 0$. This gives the deterministic contraction.

**(ii) Local-linearity perturbation.** The residual term $\eta_{\mathcal{L}_\theta} + \eta_{\mathcal{E}}$ enters as a Lipschitz perturbation of the deterministic Jacobian; standard perturbation analysis adds $O(\eta_{\mathcal{L}_\theta} + \eta_{\mathcal{E}})$ to the spectral radius.

**(iii) Stochastic Lyapunov / drift.** Under Assumption 6.2, the noise injects a bounded random walk into the iteration. A standard Foster–Lyapunov drift argument (Meyn and Tweedie, 2009) on $V(z) := \|z - z^\star\|^2$ gives, for any $\rho < 1 - \Phi_{\text{margin}}/\pi + \mathcal{O}(\eta + (MN)^{-1/2})$, $\mathbb{E}[V(z_{k+1}) \,|\, z_k] \le \rho^2 V(z_k) + b$ for some $b < \infty$. Geometric ergodicity, Wasserstein contraction, and the mixing-time estimate follow. □

# B Discussion of probing-basis sensitivity

A probe basis $V$ that misses a destabilising direction can produce a falsely positive $\Phi_{\text{margin}}$. The Hutchinson random-direction basis (Section 7.2) has the advantage that $\mathbb{E}_{v\sim\text{Unif}(\mathbb{S}^{d-1})}[\Phi_{\text{margin}}(v)]$ converges to a direction-averaged margin under enough samples; the variance bound is $\mathcal{O}(d/p)$. Hybrid bases combining task directions and random directions are recommended for safety-critical use.

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